From exponent to risk: power-law tails and the probability of large dengue epidemics in Brazil

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Abstract

Power-law tails are reported for dengue incidence and read as a verdict on forecastability: if the exponent is near two, extreme seasons cannot be anticipated. We argue that the same fit supports a more useful conclusion — that the tail prices the risk — and test it out of sample. Using weekly surveillance for the 5,571 Brazilian municipalities from 2010 to 2026, we fitted discrete power laws by maximum likelihood with Kolmogorov–Smirnov $x_{\min}$ selection at seven nested aggregation levels (47,494 fits), tested each against an exponential by likelihood ratio, and evaluated the fitted pair $(\alpha, x_{\min})$ on a rolling-origin design.

The construction decides the verdict — median exponents span 1.95 to 2.95, and the same weekly counts support a power law in 75% of municipalities pooled over a decade against 23% when cut into single seasons — but epidemic *size* survives every construction that gives it a genuine tail: $\alpha = 2.00$ in all 15 national cross-sections and 96% of 330 state-years, with $x_{\min}$ two to three orders of magnitude above the bulk.

Because $\alpha \approx 2$ fixes the whole exceedance curve, the pair is a two-parameter risk statement, and it is well calibrated: it reproduces the observed exceedance frequency over two decades, implying that 5.4–16.5% of municipalities exceed ten times $x_{\min}$ in a year and 0.3–2.7% exceed a hundred times it. The exponent is also a readout of seasonal magnitude — rank-correlated with $x_{\min}$ at 0.87 nationally and 0.65 within states, a coupling absent under simulation.

Predictively, the question worth asking is the exceedance one: lagged descriptors, with duration and peak timing as ordinary covariates, improve the Brier score over calibrated persistence in 3 of 5 origins, by up to 0.18. The scaling block adds nothing to the size forecast (0 of 5) and degrades its 95th percentile by a median 18%, because it re-encodes the lagged count. The tail is worth reporting — as a risk curve, not as a predictor.

1 Introduction

Understanding the dynamics of infectious disease epidemics has produced a very prolific literature, both in mechanistic mathematical models and in statistical ones (Siettos and Russo, 2013). The former attempt to capture the causal chain of transmission, the latter to make sense of a growing collection of incidence data.

Among the most salient results of the mechanistic tradition is the basic reproductive number R_0, which governs spreading speed and defines the epidemic threshold. On the

statistical side, most of the literature models incidence as a stochastic process, usually Poisson or negative binomial (Becker and Britton, 1999; Li et al., 2012). The central difficulty is overdispersion: surveillance counts produce far more extreme observations than any tractable count model comfortably allows.

A recurrent claim is that this overdispersion is not merely large but *scale-free*, that incidence follows a power law (Rhodes et al., 1997; Clauset et al., 2009; Neipel et al., 2020). The attraction is that a very heavy tail implies extremes that most models fail to predict — and those extremes cause most of the disease and most of the economic burden. If true, this bears directly on forecasting and on resource allocation.

The empirical record, however, is inconsistent. Reported exponents for dengue range widely, some studies find exponential rather than power-law behaviour, and the presence of a heavy tail appears to depend on the study period, the spatial unit and the outcome variable. That heterogeneity has usually been attributed to biology or geography: different places, or different years, are assumed to be governed by different regimes.

We test a narrower and more useful set of claims. We begin from a different premise than most of this literature: that a power law is not a verdict on forecastability but a *risk specification*. An exponent of two assigns a definite probability to every exceedance level, so the fitted pair $(\alpha, x_{\min})$ summarises the whole risk curve of a municipality — and it does so with two parameters, in a setting where the observables move by two orders of magnitude between years. What a heavy tail denies is that the *magnitude* of the largest event can be estimated from the variance. What it supplies is the probability that such an event occurs. Those are different problems, and a proper scoring rule keeps them apart.

What this paper does. We ask three questions in sequence, each conditional on the previous one.

First, *what is the exponent an exponent of?* Any epidemiological series can be cut into distributions in several legitimate ways — the weekly counts of one city, the annual totals of one city, the set of epidemic sizes across cities in one year, the set of epidemic durations — and these are different random variables with different supports. We fit the same estimator to the same data under seven nested constructions and report the exponent together with the diagnostic $(x_{\min})$ that makes it interpretable. Epidemic size, the quantity surveillance services actually act on, is the construction that survives.

Second, *what does the fitted pair buy?* If $(\alpha, x_{\min})$ is a risk curve, it should reproduce the empirical frequency of large seasons. We test that directly against the same cross-sections.

Third, *does the tail improve our forecasts?* We pose the question in the two forms in which it could plausibly succeed: as a classifier of whether a municipality will exceed its historical upper quartile, which is the risk statement itself; and as a quantile regression of epidemic size. Epidemic duration and peak timing enter not as objects of a scaling claim, but as ordinary covariates, since they are available before the season and may or may not help. The evaluation is a rolling-origin design with a four-rung ablation ladder and a decision rule fixed in advance.

1.1 Contributions

1. **A robust exponent for epidemic size, with the diagnostic that makes it readable.** Across seven nested constructions on one dataset, epidemic size is the only one that gives a power law at every level without collapsing into the bulk: $\alpha = 2.00$ in 15 of 15 national cross-sections and 96% of 330 state-years, with median

$x_{\min}$ of 762 and 159 cases respectively. We report $x_{\min}$ beside every exponent, which is what separates the fits that describe a tail from the fits that describe the data.

2. **The fitted pair as a calibrated risk curve.** Equation (3) turns $(\alpha, x_{\min})$ into $\Pr(S \geq s)$ for every s, and the national fits reproduce the observed exceedance frequency over two decades of epidemic size. This is a decision-relevant output that a point forecast does not supply, and it costs two parameters.

3. **A measured coupling that says what the exponent is.** Within units, α and $\log_{10} x_{\min}$ are rank-correlated at 0.65 (median over 24 states) and at 0.87 nationally, while a simulation with the true exponent fixed recovers no such association. The pair is therefore a low-variance summary of seasonal magnitude — informative about size, but not independent of the lagged case count, which is what a feature-based reading of it would require.

4. **An out-of-sample answer in the form the question can be answered.** The probability of a large season is predictable: lagged descriptors improve the Brier score over calibrated persistence in 3 of 5 origins. The scaling block adds no incremental skill to the size forecast (0 of 5) and degrades its upper quantile. The scaling description earns its place as a risk statement, not as a feature.

2 Methods

2.1 Data sources

Weekly dengue case counts were obtained from the InfoDengue/Mosqlimate platform (Araujo et al., 2025) for the period from 1 January 2010 to 22 January 2026 (the extraction date), for all Brazilian municipalities with available data. The variable used is the nowcast-adjusted estimated number of cases (`casos_est`), resampled to epidemiological weeks ending on Sunday (`W-SUN`); zero-case weeks are retained in the series but excluded from distributional fits, since the power-law family is defined on positive support.

Annual epidemic descriptors were obtained from the EpiScanner module of the same platform, which partitions each municipal series into epidemic episodes and reports, for each episode, its total size (*total cases*) and its duration in weeks (*ep_dur*). Annual cross-sections are available for the 15 epidemic years 2011–2025; the pooled all-years fit is stored separately (with a sentinel `year = 0`) and is excluded from all year counts. This yields, for each state-year and for each national year, a cross-section of municipal epidemics.

Municipal population figures were taken from the estimates distributed with the platform (`city_stats` table; 5,571 municipalities, one with a zero entry excluded). Population enters no power-law fit; it is used to compute the descriptive correlations of Section 3.1 and, as $\log_{10}$ population, as one predictor in the forecasting models of Section 2.7.

2.2 Seven aggregation levels

The unit of analysis is the *set of positive observations* passed to the estimator. Table 1 defines the seven constructions used. The first three treat the individual municipality as the sampling unit and vary the temporal window; the last four treat a cross-section of municipalities within a period as the sampling unit.

Levels L1 and L2 differ only in the temporal window over which the weekly counts are pooled: L1 uses the complete available series for a municipality (January 2010 to January 2026), whereas L2 restricts to a single epidemic year (at most 52 weeks). Both sample

Table 1: The seven aggregation levels. "Unit" is the sampling unit, i.e. what each observation in the fitted sample represents.

ID	Unit	Variable	Source table
L1	city	weekly cases, full series	`powerlaw_fits`
L2	city-year	weekly cases, one season	`powerlaw_fits_yearly`
L3	city	annual total, across years	`powerlaw_fits_total_cases`
L4	city $\subset$ state-year	epidemic size	`episcanner_state_fits`
L5	city $\subset$ state-year	epidemic duration	`episcanner_state_fits`
L6	city $\subset$ national-year	epidemic size	`episcanner_fits`
L7	city $\subset$ national-year	epidemic duration	`episcanner_fits`

sizes therefore differ by roughly an order of magnitude, which is why we compare shares rather than raw counts. Levels L4–L7 differ only in the spatial scope of the cross-section, not in the variable being measured. This nesting is deliberate: it allows each contrast in Section 3 to be read as the effect of a single change in construction. Figure 1 shows the scheme.

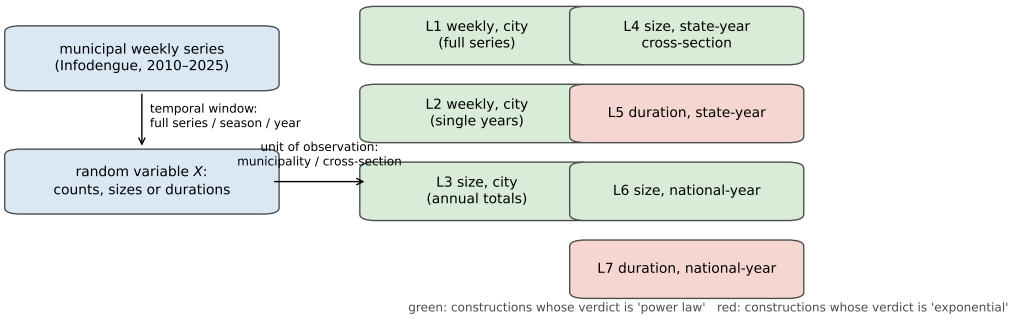


Figure 1: The seven aggregation levels as nested constructions on a single dataset. Starting from the same municipal weekly series (top), the analyst chooses a temporal window — the full series, one season, or one year — and a unit of observation — either the municipality itself, or the cross-section of municipalities within a period — and obtains random variables that differ in support, variance and tail behaviour. L1 and L2 differ only in the window; L3 aggregates each municipality over years; L4–L7 differ only in the spatial scope of the cross-section (state-year versus national-year). Every contrast reported in Section 3 is a single step in this scheme, so a difference in the inferred regime is attributable to the construction rather than to a change of data source, period or software.

2.3 Discrete power-law estimation

For each sample we fit a discrete power law by maximum likelihood following Clauset et al. (2009). The fitted density for integer $x \geq x_{\min}$ is

$$p(x) = \frac{x^{-\alpha}}{\zeta(\alpha, x_{\min})}, \tag{1}$$

where $\zeta(\alpha, x_{\min})$ is the generalised (Hurwitz) zeta function, which normalises the discrete mass. The exponent α is estimated by numerical maximisation of the likelihood,

and the lower bound $x_{\min}$ is selected jointly by minimising the Kolmogorov–Smirnov distance between the empirical and fitted cumulative distributions over all candidate values of $x_{\min}$. Because the data are counts, the discrete formulation is used throughout (`discrete=True`); the continuous approximation is not employed at any level.

One property of the estimator has to be stated because it bears on the duration results. The numerical fit constrains α to $(1, 3]$: the upper limit of the optimiser's default parameter range is 3, so an estimate cannot exceed it and a sample whose likelihood is maximised beyond that value returns $\alpha = 3$ rather than its unconstrained maximum. The bound is immaterial where the fitted exponents lie well inside the range. At the cross-sectional size levels the largest value is 2.99 (L4) and 2.27 (L6); at the two weekly levels it is reached by 2 and 5 fits respectively, under 0.05% of each. It binds materially only for durations, where the upper quartile of the 330 state-year fits lies above 2.92 and 11 of them (3.3%) sit at the ceiling. Those fits should be read as *at least* 3, not as 3, and a genuinely scale-free body would not systematically exhaust the upper bound of the estimator.

The choice of $x_{\min}$ is not a nuisance parameter. It determines which fraction of the data the model actually describes, and therefore whether the fitted exponent refers to a tail at all. We therefore retain $x_{\min}$ for every fit and report its median and the proportion of fits with $x_{\min} \leq 5$ in each aggregation level.

2.4 Model comparison

A power law can always be fitted; the question is whether it describes the data better than a plausible alternative. We compare each power-law fit against an exponential distribution, which is the natural null for epidemiological counts because it corresponds to a constant hazard of termination and produces a light tail. The comparison follows the likelihood-ratio procedure of Clauset et al. (2009), as implemented in the `powerlaw` package with `normalized_ratio=True`.

$$R = \sum_{i=1}^{n} \left[\ln p_{\mathrm{PL}}(x_i) - \ln p_{\exp}(x_i) \right], \tag{2}$$

normalised so that its magnitude is comparable across samples of different size. The sign of R indicates the preferred model ($R > 0$ favours the power law, $R < 0$ favours the exponential) and its associated p-value quantifies whether the sign is distinguishable from sampling noise.

We adopt a deliberately conservative reporting convention. A level is described as *power-law supported* only when $R > 0$ and $p < 0.05$; as *exponential supported* only when $R < 0$ and $p < 0.05$; and as *inconclusive* otherwise. This prevents the common error of counting a positive but non-significant R as evidence for scaling.

2.5 Identification diagnostic

We flag a fit as *tail-informative* when $x_{\min}$ exceeds the bulk of the observed distribution, operationally $x_{\min} > 5$ cases for weekly counts and $x_{\min}$ above the median observed value for the aggregate levels. Where $x_{\min} \leq 5$, the fitted power law is being asked to describe the entire empirical distribution, including the small-count regime where the discrete power law is not a plausible generative model (case counts of 1 or 2 arise from reporting mechanics, not from scale-free branching). Such fits are retained in the database but are excluded from substantive interpretation, and we report the proportion explicitly.

2.6 From description to prediction

A fitted power law is not merely a description of a histogram; it is a statement about exceedance probabilities. For integer $s \geq x_{\min}$,

$$\Pr(S \geq s) \;=\; \frac{\zeta(\alpha, s)}{\zeta(\alpha, x_{\min})}, \tag{3}$$

so a single pair $(\alpha, x_{\min})$ parameterises the probability that a municipality experiences an epidemic of at least any given size. Four consequences of Equation (3) shape the forecasting design.

First, the exponent summarises *risk* rather than *activity*. Two municipalities with the same expected annual incidence but different α face different probabilities of a catastrophic season, and that difference is invisible to the mean. The fitted models quantify the claim rather than merely asserting it: using the 15 national size fits, they imply that between 5.4% and 16.5% of municipalities exceed ten times $x_{\min}$ in a given year, and between 0.3% and 2.7% exceed one hundred times $x_{\min}$. Those are falsifiable implications of the description, and Figure 9 confronts them with the empirical exceedance frequencies.

Second, α is a low-variance quantity *relative to its target*. Across the national cross-sections it takes values between 1.78 and 2.27, and it is essentially unrelated to city size ($r = -0.065$). Annual case counts over the same period vary by two orders of magnitude. The dispersion of $\hat{\alpha}$ is not zero — it exceeds sampling noise by about a factor of five in standard deviation, and it co-varies with the selected $x_{\min}$ (Section 3.3) — but it is small along precisely the dimension in which the target is volatile. That is the specific reason to carry α into the predictive models rather than treating it as a descriptive curiosity, and the same coupling is the reason the ablation, not the correlation, has to decide the question.

Third, $\alpha \approx 2$ has a direct methodological consequence for evaluation. The second moment of a power law with $\alpha \leq 2$ does not exist in the continuous limit, and at $\alpha \approx 2$ the sample mean is unstable, so squared-error losses are decided by the few largest outbreaks. A model that is good everywhere except in the tail can therefore look excellent or terrible depending on which years fall in the test set. We consequently evaluate on quantile (pinball) losses and on the classification of tail events, and treat RMSE as a secondary, tail-dominated diagnostic.

Finally, the predictive hypothesis is stated in a form that can fail. If $(\alpha, x_{\min})$ carries information about the coming season beyond what lagged counts already encode, a model containing them should beat an otherwise identical model without them, out of sample. If instead the exponent is close to a deterministic function of the lower moments of the same series, the two models should be indistinguishable — and the descriptive result would then carry no forecasting content. Section 2.10 specifies the comparison that decides between these outcomes, and Section 2.11 lists the three ways in which such a comparison could be faked and how each is prevented.

2.7 Predictive targets and feature construction

The forecasting problem is posed at the level of an administrative unit (municipality or state) and a target year t: every predictor is measured in year $t-1$ or earlier, and the target is a property of the epidemic season of year t. Three targets are modelled, corresponding to the descriptors that surveillance services actually act on: epidemic size S_t (total cases in the annual episode), epidemic duration D_t (weeks), and peak timing P_t (week of the seasonal maximum).

Predictors fall into six blocks, listed in Table 2: lagged epidemic metrics; the scaling descriptors $(\alpha_{S,t-1}, x_{\min,S,t-1})$ from the power-law fit to the previous season; the historical trend $\dot{\alpha}_{t-1}$, defined as the ordinary least-squares slope of α_S over all years strictly before t and set to zero where fewer than two such years exist; the epidemiological driver $R_{0,t-1}$ reported by the EpiScanner decomposition; $\log_{10}$ population; and eight quarterly climate aggregates from year $t-1$, namely mean minimum temperature and mean minimum humidity in each of Q1–Q4. Units lacking the mandatory lags (no previous season) are dropped rather than imputed; the optional $\dot{\alpha}$ is set to zero when undefined.

Table 2: Predictor blocks and the sources from which they are computed. All predictors refer to year $t-1$ or earlier. The scaling block is the object of the ablation in Section 2.10.

Block	Variables	Source
Lagged metrics	S_{t-1}, D_{t-1}, P_{t-1}	`episcanner`
Scaling descriptors	$\alpha_{S,t-1}$, $x_{\min,S,t-1}$	`powerlaw_results.db`
Scaling trend	$\dot{\alpha}_{t-1}$	derived (OLS on α_S)
Epidemiological	$R_{0,t-1}$	`episcanner`
Demographic	$\log_{10}(\text{pop})$	`city_stats`
Climate	$\bar{T}_{Q1..Q4}$, $\bar{H}_{Q1..Q4}$	Infodengue (`tempmin`, `umidmin`)

2.8 Regression models

Epidemic size is modelled on the log scale, $\log(1 + S_t)$, because the target spans four orders of magnitude and a model fitted on the raw scale would in practice be fitted to the largest municipalities only; predictions are returned to the case scale with $\exp(\cdot) - 1$. That back-transformation is biased downwards by Jensen's inequality, so a multiplicative factor $c = \overline{S}_{\text{train}} / \overline{(\exp \hat{y} - 1)}_{\text{train}}$ is estimated on the training data and applied to the point predictions. The correction is a mean-matching device, not a distributional one, and it is deliberately not applied to the quantile models, whose losses are computed on the back-transformed quantiles themselves. Duration and peak week are modelled on their natural scale.

Three estimators are used. (i) A random forest of 150 bootstrapped trees (Breiman, 2001), which additionally provides an out-of-bag R^2 reported as an in-sample fit diagnostic. (ii) A LightGBM gradient-boosted model with quantile objective at level τ (Ke et al., 2017; Koenker and Bassett, 1978), with 300 trees, learning rate 0.05, 16 leaves and a minimum of 5 observations per leaf. (iii) A histogram gradient-boosting model, also with quantile loss, 300 iterations, learning rate 0.05 and maximum depth 5. The quantile models are fitted on the grid $\tau \in \{0.05, 0.10, \ldots, 0.95\}$, which yields a full predictive distribution per unit and year rather than a point forecast. All stochastic components use `random_state=42`.

2.9 Classification of large epidemics

The second predictive task is posed as a decision problem rather than an estimation problem: will unit i experience, in year t, a season larger than the p-th percentile of its own historical distribution? The target is

$$Y_{i,t} \;=\; \mathbb{I}\big[\, S_{i,t} > Q_p\big(\{S_{i,k}\}_{k<t}\big) \,\big]\,, \tag{4}$$

with $p = 0.75$ by default. The threshold is a *within-fold* quantity: it is recomputed from the training years of each fold and each split, never from the full sample, so that the label of a test observation cannot depend on information unavailable to the model. The same discipline governs the class-imbalance weight, set to the negative-to-positive ratio of the training fold.

Two gradient-boosted classifiers are used, XGBoost (Chen and Guestrin, 2016) and LightGBM, with imbalance handled by `scale_pos_weight` and `is_unbalance` respectively. Validation is an expanding-window split defined on the year index. Fold j trains on every unit-year whose year is earlier than v_j and validates on year v_j alone, where the v_j are the $k = \min\{5, n_{\text{years}} - 2\}$ most recent training years: a fold exists only once at least two earlier years are available, and the number of folds is capped at five. Because the folds are a function of the year vector and not of row position, they are invariant to the order of the rows in the frame. Within each fold the percentile threshold that defines the label, the class-imbalance weight and the median imputer are all recomputed from that fold's training years, so no quantity used to label or to weight a validation year can have been influenced by it; a per-fold audit table (training window, held-out year, fold sizes, threshold, weight) is retained with the results.

This distinction is not cosmetic, and it is the one step of this pipeline whose failure is silent. An implementation that applies a contiguous-index split to a frame sorted by unit rather than by year produces a grouped-by-municipality split: each fold trains on some municipalities over the whole calendar and validates on others over the same calendar, so validation contains no year that is absent from training. On the training panel used here — 32,824 rows, 4,725 municipalities, 2010–2023 — all five folds of such a split train on 2010–2023 and validate on 2010–2023, and 100% of validation rows carry a year already present in training. The quantity being estimated is then transportability to new municipalities, not temporal generalisation, and the estimate depends on an undeclared sort order: merely reordering the rows of a synthetic benchmark moved its cross-validated AUC from 0.813 to 0.835, whereas the year-indexed split returns the same folds and the same AUC under every permutation of the rows. The trap is narrower but not absent when the frame happens to be year-sorted, because contiguous blocks then still cut through calendar years. The splitter therefore takes the year vector explicitly, and `assert_forward_in_time` raises rather than warns when any validation year is not strictly later than the training years of its fold. The decision threshold applied to the predicted probability is chosen by Youden's J statistic (Youden, 1950), computed on the out-of-fold predictions rather than on the training fit, and then held fixed for the test set. We report the area under the ROC curve, the area under the precision–recall curve (the more informative summary under class imbalance), the Brier score (Gneiting and Raftery, 2007), the F_1 score at the Youden threshold and a calibration curve, and we test differences in AUC between nested feature sets by the method of DeLong et al. (1988). Feature attributions are obtained from SHAP values (Lundberg and Lee, 2017).

2.10 Evaluation protocol and ablation ladder

A single temporal split cannot distinguish a feature that helps from a feature that happened to help once. We therefore adopt a rolling-origin design: for each forecast origin $y \in \{2021, \ldots, 2025\}$ the model is trained on all observations with year $< y$ and evaluated on year y, and the per-origin losses are pooled. This gives five honest out-of-sample years and allows skill to be read as a function of the history available to the model.

Within each origin, the incremental value of the scaling description is isolated by the ablation ladder of Table 3. Each rung adds one block of predictors to the previous one,

so a contrast between consecutive rungs is attributable to the block that was added. The rung of substantive interest is M2 versus M3, which differ *only* in the presence of $(\alpha_{S,t-1}, x_{\min,S,t-1}, \dot{\alpha}_{t-1})$.

Table 3: The ablation ladder. M0 is the benchmark that any claim of skill must beat; successive rungs add one predictor block so that consecutive contrasts are attributable to that block. The M2→M3 step isolates the contribution of the power-law scaling descriptors.

Rung	Name	Predictors
M0	Persistence	$\hat{S}_t = S_{t-1}$, $\hat{D}_t = D_{t-1}$; historical median for P_t; training base rate for the classifier
M1	Lagged metrics	M0 predictors + P_{t-1}, $R_{0,t-1}$, $\log_{10}(\text{pop})$
M2	+ climate	M1 + quarterly $\bar{T}$, $\bar{H}$ from year $t-1$
M3	+ scaling	M2 + $\alpha_{S,t-1}$, $x_{\min,S,t-1}$, $\dot{\alpha}_{t-1}$

Operationally, M0 forecast values are scaled by the empirical τ-quantile of the within-unit year-on-year ratio, estimated on the training years, so that the same benchmark can be evaluated at every quantile level rather than only at the median. At $\tau = 0.5$ the scale factor is close to unity and M0 reduces to plain persistence; at the other levels this is the honest analogue of persistence, not a straw man.

M0 is deliberately naive, because a benchmark that is already sophisticated manufactures skill for whatever is added to it. Skill is reported as

$$\text{SS} \;=\; 1 - \frac{L(\text{model})}{L(\text{M0})}, \tag{5}$$

on the primary loss of each task — the pinball loss at $\tau = 0.5$ for each of the three regression targets, and the Brier score for the classifier, both proper scoring rules (Gneiting and Raftery, 2007). Secondary losses are MAE and RMSE of the point forecasts, the mean absolute scaled error of Hyndman and Koehler (2006) to make errors comparable across units of very different size, and the pinball loss at each τ on the grid.

Because units differ by orders of magnitude in scale, all model-versus-model comparisons are made on paired loss differentials computed within the same unit and year; their uncertainty is estimated by a bootstrap that resamples *units* rather than observations, preserving the within-unit correlation across years, with 2,000 replicates. For the pooled comparison across origins we additionally report the Diebold–Mariano statistic (Diebold and Mariano, 1995) with a Newey–West correction, since the loss series are serially correlated by construction.

The decision rule is fixed in advance. The scaling block is judged to add predictive value only if M3 improves on M2 on the primary loss with a 95% bootstrap interval for the paired difference that excludes zero, in the same direction, for at least three of the five forecast origins. This is stricter than a single significant pooled test, and deliberately so: a pooled test can be carried by one anomalous season, which is exactly the failure mode that a heavy-tailed target invites.

2.11 Leakage controls

Four sources of leakage are structurally possible in this design, and each is closed explicitly.

1. **Contemporaneous scaling descriptors.** $\alpha_{S,t}$ is estimated from the weekly counts of season t, which are part of the target. Only $\alpha_{S,t-1}$ enters as a predictor, and $\dot{\alpha}_{t-1}$ is computed exclusively from years strictly before t.

2. **The classification threshold.** Q_p is computed inside each fold from training years only. Computing it once on the full sample would leak the distribution of the test years into the labels of the training years, inflating apparent skill by construction.

3. **Cross-sectional information.** The cross-section of municipalities within year t is an outcome of season t and is therefore never a predictor; the scaling descriptors always come from the previous season's cross-section.

4. **The validation split itself.** Folds are defined on the year index (Section 2.9), so no fold trains on a year at or after the year it validates on. This is the only one of the four channels that fails silently in a plausible implementation — every number still looks like a respectable cross-validation score — which is why it is asserted in code rather than merely documented.

2.12 Software and reproducibility

All fits are stored in a single SQLite database (`powerlaw_results.db`) committed to the project repository, together with the fitting code (`fitpl.py`) and the cached parquet extracts of every model result. Every number in this manuscript is derived from that database by the queries in Appendix A; no statistic is transcribed by hand. The full pipeline is therefore re-runnable, and a reader can confirm or contradict each figure with a single SQL statement.

3 Results

Table 4 summarises the 47,494 fits across the seven aggregation levels. We read it in the order of the questions above: what the exponent is an exponent of, what the fitted pair buys as a risk statement, and what it adds to a forecast. Figure 2 shows the estimator's output at each level, Figure 3 the corresponding verdicts, and Figure 4 the diagnostic that separates the constructions worth interpreting from those that are not.

3.1 The aggregation level decides the verdict

One estimator, one body of surveillance data, seven ways of cutting it, and verdicts ranging from unanimous support for a power law (L6, 15 of 15 years) to unanimous rejection in favour of an exponential (L7). Median exponents span 1.95 to 2.95 and selected lower bounds run from a single case to about 1.5×10^5, five orders of magnitude. Nothing distinguished the analyses except the construction.

Two things disqualify half of the table as a basis for a scaling claim, and both are visible without a new experiment. Where the estimated lower bound collapses into the bulk, the fit describes the whole distribution and the tail language is vacuous: 77% of L1 and 81% of L2 fits have $x_{\min} \leq 5$, i.e. the model is estimated on counts of one and two cases. And where the window changes, the verdict changes with it: pooling the same weekly counts over a decade instead of a single season drops the proportion of fits favouring a power law from 75% to 23% and raises significant support for the exponential from 4% to 21%. Not a transmission parameter differs between those two numbers (Figure 6). What the difference looks like for individual fits is in Figure 5.

Table 4: Power-law fits by aggregation level. α median with interquartile range; $x_{\min}$ median; $\mathrm{PL}_{\mathrm{pref}}$ = proportion with $R > 0$; $\mathrm{PL}_{\mathrm{sig}}$ = proportion with $R > 0$ and $p < 0.05$; $\mathrm{Exp}_{\mathrm{sig}}$ = proportion with $R < 0$ and $p < 0.05$. Shares are percentages. The levels that survive the $x_{\min}$ diagnostic *and* the model comparison are the cross-sectional size constructions L4 and L6, which are the object of the rest of the paper.

Level	n	α	IQR	$x_{\min}$	$\mathrm{PL}_{\mathrm{pref}}$	$\mathrm{PL}_{\mathrm{sig}}$	$\mathrm{Exp}_{\mathrm{sig}}$
L1 weekly, city	5,244	1.95	1.72–2.29	2	75	45	4
L2 weekly, city-year	40,616	2.06	1.72–2.45	2	23	3	21
L3 annual total, city	944	2.09	1.80–2.46	341	51	4	4
L4 size, state-year	330	1.97	1.79–2.14	159	96	52	0
L5 duration, state-year	330	2.80	2.52–2.92	17	0	0	95
L6 size, national-year	15	2.00	1.92–2.11	762	100	100	0
L7 duration, national-year	15	2.95	2.90–2.96	18	0	0	100

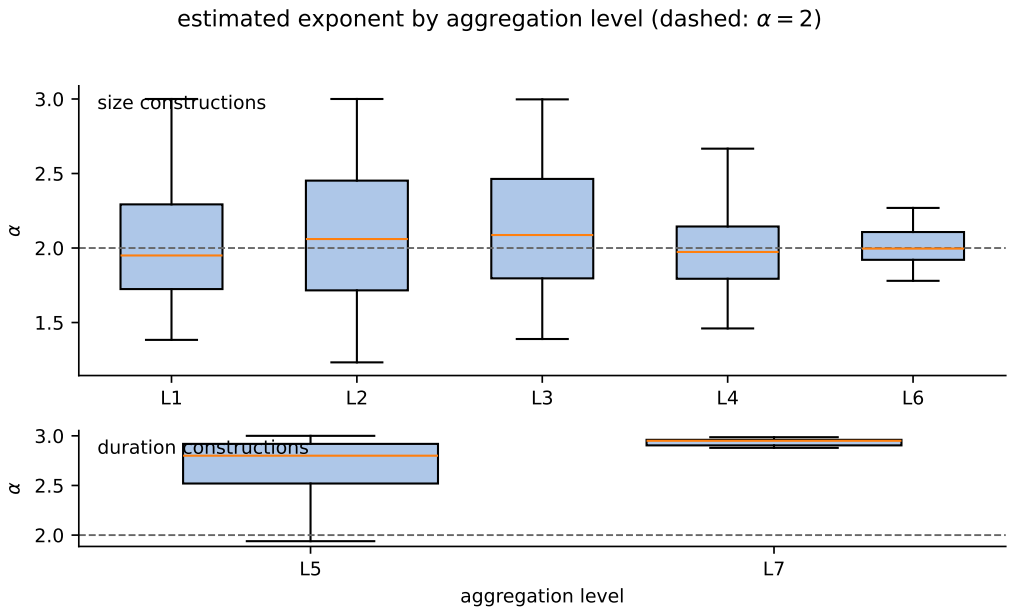


Figure 2: Distribution of the estimated exponent α at each aggregation level. Boxes give the median and interquartile range across fits, whiskers extend to 1.5 times the IQR, and the dashed line marks $\alpha = 2$. The upper panel groups the size constructions (L1–L4, L6); the lower panel groups the duration constructions (L5, L7), which are shown here for completeness and are not used further. Size distributions concentrate near $\alpha \approx 2$ at every level; durations pile up near $\alpha \approx 2.9$, the value at which a discrete power law approximates a truncated or exponential body, so the estimator returns a near-constant exponent whatever the underlying process.

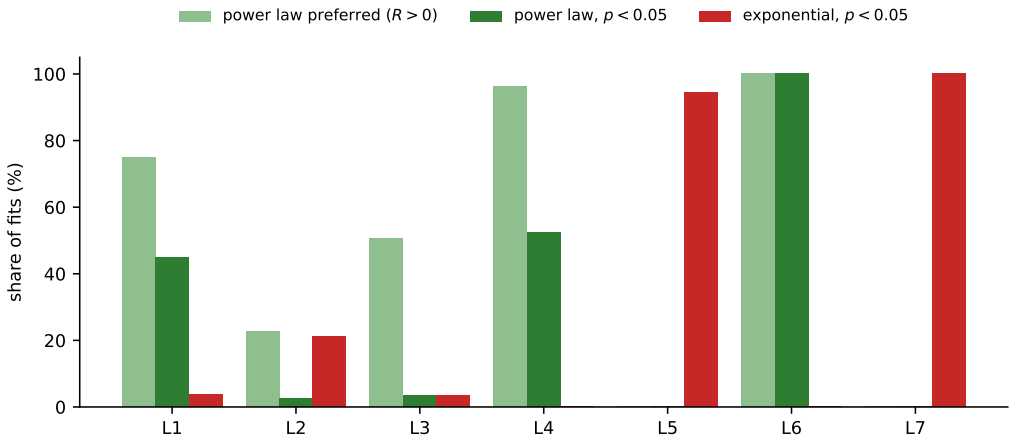


Figure 3: Model comparison by aggregation level. For each level, the proportion of fits in which the power law is preferred ($R > 0$), the proportion in which it is preferred *and* significant at $p < 0.05$, and the proportion in which the exponential is significant ($R < 0$ and $p < 0.05$). Levels are ordered from the individual-series constructions (left) to the cross-sectional ones (right). The two weekly levels produce apparent power-law support that the $x_{\min}$ diagnostic removes; the cross-sectional size levels are the only construction whose support survives both tests.

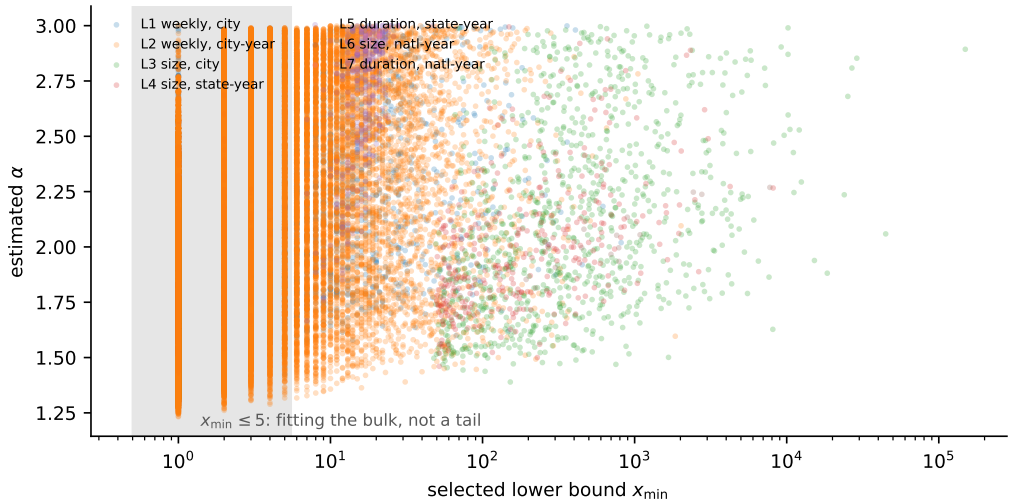


Figure 4: Identification diagnostic. Estimated exponent α against the selected lower bound $x_{\min}$ (log scale), one point per fit, coloured by aggregation level. The shaded band marks $x_{\min} \leq 5$, where the fitted power law is asked to describe the bulk of the distribution rather than a tail. The two weekly levels sit almost entirely inside that band, while the cross-sectional levels sit two to three orders of magnitude to its right.

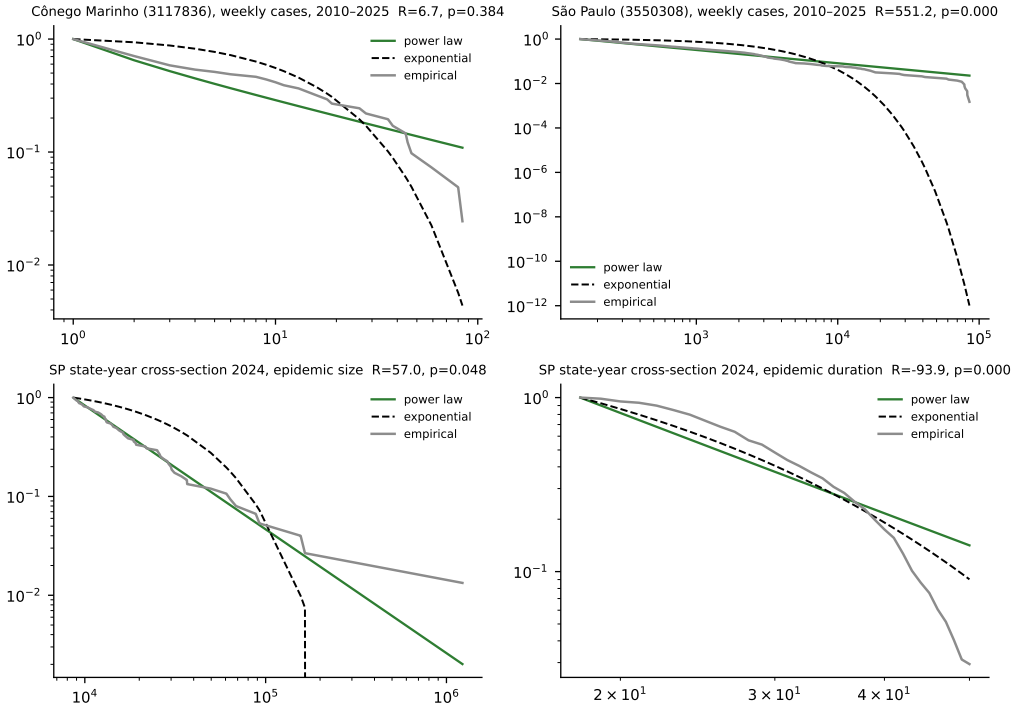


Figure 5: Individual fits across the aggregation levels, each with the power-law (solid) and exponential (dashed) alternatives superimposed. (a) Weekly counts of a small municipality, where the selected $x_{\min}$ collapses to 2 and the "power law" is fitted to the entire distribution; (b) weekly counts of a large state capital; (c) a state-year cross-section of epidemic sizes, where $x_{\min}$ lies in the hundreds of cases; (d) the corresponding cross-section of durations. Panels (a) and (c) are produced by the same estimator applied to different objects.

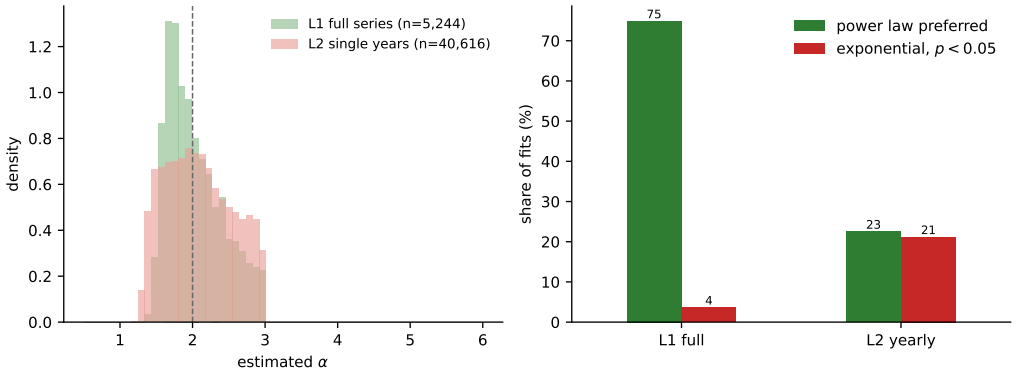


Figure 6: The window flips the verdict on identical data. Left, density of the estimated exponent from the full municipal series (L1) and from the same series cut into single seasons (L2). Right, the corresponding verdicts: the share of fits favouring the power law falls from 75% to 23%, while the share significantly favouring the exponential rises from 4% to 21%.

3.2 Epidemic size is power-law with an exponent of two

The construction that survives both tests is the one that matters operationally: the distribution of epidemic *size* across units within a year. At the national cross-section (L6) all 15 annual fits favour the power law and all are significant, with $\alpha = 2.00$ (IQR 1.92–2.11) and $x_{\min} = 762$ cases. The exponent is remarkably stable, ranging only from 1.78 to 2.27 over fifteen years, in a period in which annual case counts vary by two orders of magnitude. At state level (L4) the picture is consistent but weaker in significance, as expected from smaller cross-sections: the power law is preferred in 96% of the 330 state-year fits, significant in 52%, with median $\alpha = 1.97$ and median $x_{\min} = 159$ cases.

This is the object the rest of the paper works with, and it is worth being precise about why. The median lower bound of 762 cases at national level and 159 at state level places the fit well above the bulk of a distribution whose observations span four orders of magnitude: what is being described is the upper region where the risk sits, not the body of small seasons. The empirical complementary cumulative distribution of Figure 7 shows the same thing graphically — a straight line over two decades above $x_{\min}$.

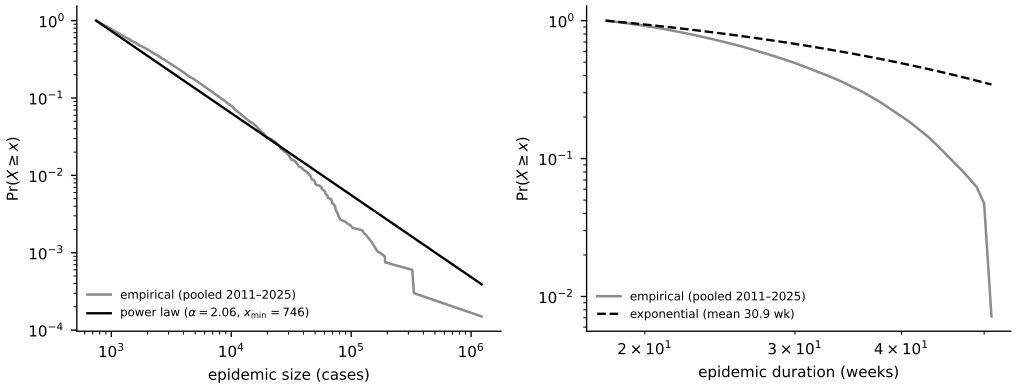


Figure 7: Empirical complementary cumulative distributions at the national cross-section, pooled over the 15 epidemic years, with the selected model superimposed. Left, epidemic size on log–log axes with the fitted power law: the curve is close to linear over two decades above $x_{\min} = 762$ cases. Right, epidemic duration on the same axes, shown for contrast: the curve bends downwards and is captured by an exponential. A straight line on the left is the visual form of the claim this paper goes on to use.

3.3 What the exponent is: a readout of the scale of the season

The fitted pair is not a pair of independent coefficients, and the direction of that dependence determines how the description can be used. Table 5 collects the association between the two estimates at the levels where it can be measured, together with the association recovered from simulated data in which the true exponent was held fixed.

The association is strong at national level and strong *within* states over time — positive in 88% of the 24 states with at least six years of data, and significantly positive in 67% — but weak between states. It is therefore a within-unit, over-time relationship, not a contrast between places. The control recovers no such association, so the estimator does not manufacture it: it places the sampling dispersion of $\hat{\alpha}$ at 0.03 (40 replicates, samples of 800–4,000 points, true exponent fixed at 2.0), against an observed standard deviation

Table 5: Association between the estimated exponent and the selected lower bound. Rows 1–3 are the surveillance data; the last row is a control in which data were generated from a power law with true $\alpha = 2.0$ while the true lower bound was varied over the same range, so that any association recovered there is an artefact of the estimator. Values are Spearman rank correlations between α and $\log_{10} x_{\min}$.

Stratum	Sample	ρ	n
National cross-section, year to year	15 years	0.87	15
Within state, year to year	median of 24 states	0.65	–
Between states	state means	0.22	26
Simulated, α fixed at 2.0	control	−0.20	40

of 0.154 across the 15 national fits. The year-to-year movement in α is real, roughly five times the sampling noise in standard deviation, even though it remains small next to the variation in the case counts themselves.

Read as description, this is the most informative single result about $(\alpha, x_{\min})$ in the paper. Because $x_{\min}$ is measured in cases, it tracks the magnitude of the season: years in which municipalities produce large epidemics place the cut higher, and the exponent moves with it. The pair is therefore a two-number summary of how large a season was — one that is available for every municipality, is far less volatile than the case count it summarises, and is stable enough to be compared across years (Figure 8). Read as a feature set, the same fact is a warning: an exponent that co-varies with the scale of the season is, to that extent, a re-encoding of S_{t-1}, which any model of size already carries. Both readings are correct, and they predict different things about the ablation in Section 3.6.

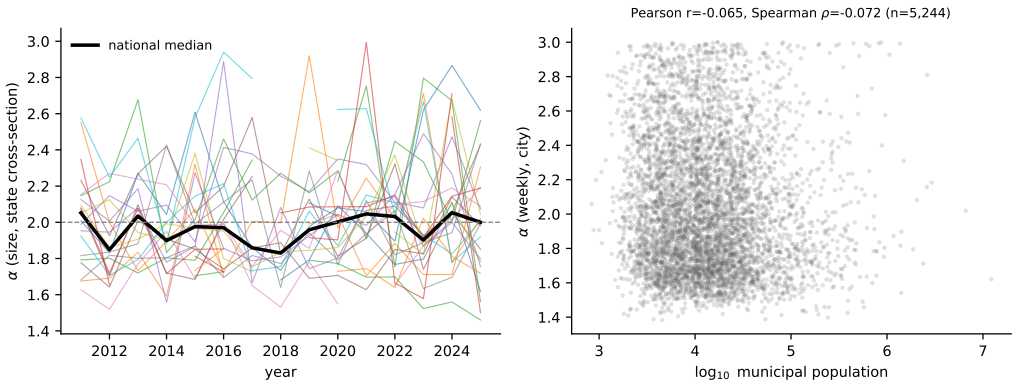


Figure 8: Stability of the exponent. (a) Estimated α for epidemic size by year, one line per state, at the state cross-section: the lines are flat, and the spread between states is far smaller than the year-to-year variation in case counts. (b) α against $\log_{10}$ municipal population, which shows no association (Pearson $r = -0.065$, Spearman -0.072, $n =$ 5,244). Panel (a) is the empirical reason the exponent can serve as a compact descriptor of a season: it varies slowly along exactly the axis in which the observables do not.

The exponent is essentially unrelated to city size ($\mathrm{corr}(\alpha, \log_{10}\text{population}) = -0.065$, Spearman -0.072; $n = 5{,}244$) and only weakly related to incidence (Pearson -0.136,

Spearman -0.248; $n = 4{,}663$). Larger or more heavily burdened municipalities are not in a different scaling regime: the exponent is a property of the construction, and within a unit of the scale of the season.

3.4 The fitted law as a risk curve

Equation (3) makes the description operational. A power law with exponent α and lower bound $x_{\min}$ assigns $\Pr(S \geq s) = \zeta(\alpha, s)/\zeta(\alpha, x_{\min})$ for every level s, so the fitted pair is a complete statement about how often extreme seasons occur. It requires no fitted case count, no covariate, and no training period: it is available from the current cross-section alone.

Figure 9 confronts the exceedance curve implied by each national fit with the empirical exceedance frequency of the same cross-section. The agreement holds over roughly two decades of epidemic size. Numerically, the 15 national fits imply that between 5.4% and 16.5% of municipalities exceed ten times $x_{\min}$ in a given year, and between 0.3% and 2.7% exceed a hundred times it. Those are statements a surveillance service can act on and a subsequent season can falsify.

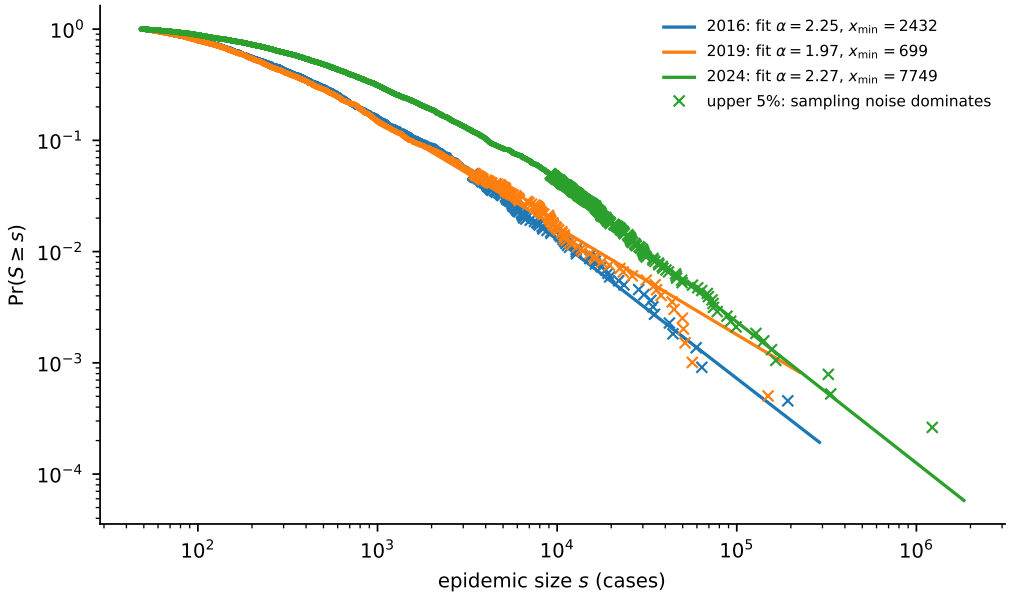


Figure 9: The fitted power law as a probabilistic statement about exceedance. For each national cross-section, curves give $\Pr(S \geq s) = \zeta(\alpha, s)/\zeta(\alpha, x_{\min})$ evaluated at the fitted $(\alpha, x_{\min})$, and points give the empirical exceedance frequency of the same cross-section. Agreement across two decades of epidemic size is what licenses the use of $(\alpha, x_{\min})$ as a compact statement about a municipality's risk of an extreme season. Where the curves separate from the points in the extreme upper tail, the sample is too small to distinguish model error from sampling noise; those regions are marked.

This is the sense in which the description is predictive before any machine learning is involved: α is not a summary of central tendency but a summary of how much of the risk sits in the tail, and it is available for every municipality even when that municipality has

never experienced an extreme season. The familiar inference — that a divergent second moment makes extreme seasons unpredictable — is a statement about squared-error loss, not about information. Prediction of the tail event and prediction of its size are different problems.

3.5 Predicting the probability of a large epidemic

The exceedance formulation has a direct predictive counterpart: rather than asking how large the season will be, ask whether it will fall in the upper quartile of the unit's own history. This is the risk statement itself, and it is the natural place to look for value in the scaling descriptors, since a tail exponent is by construction a statement about upper-tail probabilities.

It is also the one place where the answer is positive. Table 6 reports the paired Brier-score differentials across the five forecast origins. Lagged epidemic descriptors — which include the previous season's duration and peak timing as ordinary covariates, alongside size, R_0 and population — improve on calibrated persistence with a 95% bootstrap interval excluding zero in 3 of the 5 origins, by up to 0.175 in absolute Brier score. Climate adds nothing (0 of 5), and the scaling block adds nothing (0 of 5), with point estimates close to zero.

Table 6: The large-season classifier, state level, five forecast origins 2021–2025. The target is whether a unit exceeds the 75th percentile of its own history; the loss is the Brier score. Positive values mean the later rung has the lower Brier score. The decision rule of Section 2.10 requires a same-direction improvement with a 95% unit-bootstrap interval excluding zero in at least 3 of 5 origins.

Contrast	ΔBrier, median	Origins $\Delta > 0$	Origins CI excl. 0	Verdict
M1−M0 (lagged metrics)	+0.134	4/5	**3/5**	**supported**
M2−M1 (climate)	−0.001	2/5	0/5	not supported
M3−M2 (scaling)	+0.008	3/5	0/5	not supported

Per origin, the M1−M0 differentials are −0.047, +0.074, +0.134, +0.175 and +0.159 for 2021 to 2025 — negative in the first origin and increasingly positive thereafter, which is what one expects from a model that needs history to be trained on. The scaling block's differentials are −0.009, −0.001, +0.008, +0.036 and +0.010: three of five have the right sign and none has an interval excluding zero. Whatever the exponent contributes to the probability of a large season, it is not detectable against a model that already knows how large the previous season was.

Figures 10 and 11 report discrimination, calibration and attribution for the classifier. The calibration panel is the one to read: a classifier that is useful for planning must track the diagonal in the upper deciles, not merely rank well on average.

3.6 Predicting epidemic size

The harder question is whether the tail helps forecast the magnitude of a season, which is what a health secretariat orders beds against. Here the answer is negative, and the reason is the coupling measured in Section 3.3.

Table 7 reports the ladder over the five origins at state level. Against calibrated persistence, the lagged-metrics rung M1 carries essentially nothing for size (median skill 0.10, range −0.59 to 0.71, positive in 3 of 5 origins and 2 of 5 with an interval excluding

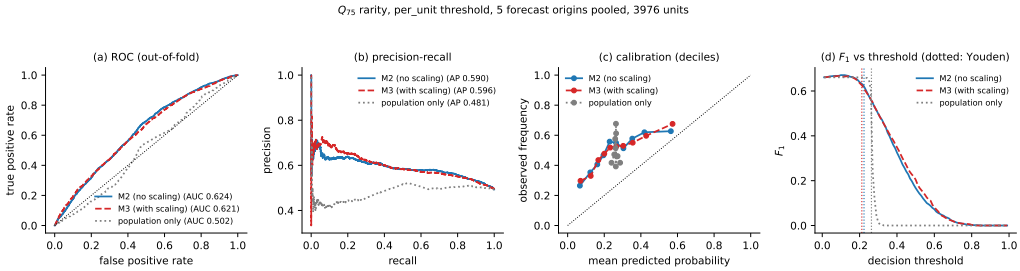


Figure 10: Classification of large epidemics. (a) ROC curves; (b) precision–recall curves, the more informative summary when the positive class is the minority; (c) calibration of the predicted probability of exceeding the 75th percentile; (d) F_1 as a function of the decision threshold, with the Youden point marked. All curves are drawn from out-of-fold predictions so that they are honest. Solid lines denote M2 (without scaling descriptors) and dashed lines M3 (with them).

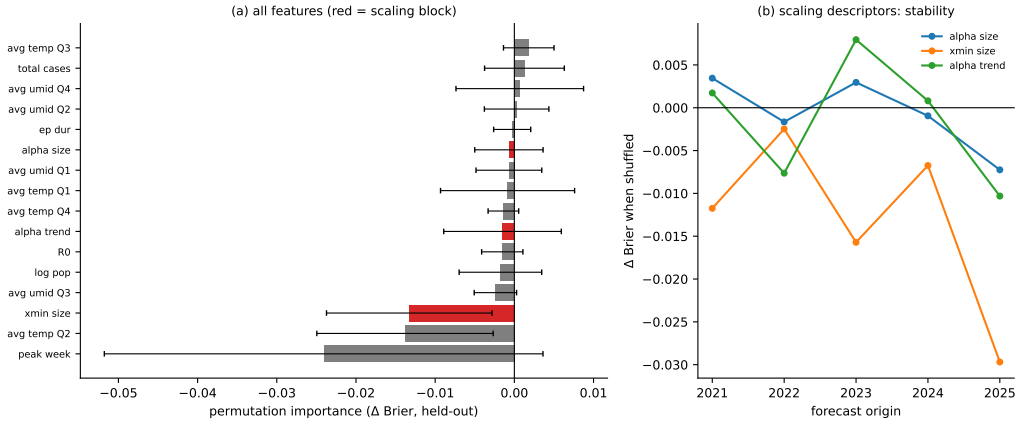


Figure 11: Where the scaling descriptors rank. Mean absolute SHAP values for the large-season classifier, pooled over forecast origins, with the power-law block highlighted. The left panel ranks all features; the right panel isolates α_{t-1}, $x_{\min,t-1}$ and $\dot{\alpha}_{t-1}$ and shows the direction of their effect on the predicted probability.

zero), and successive rungs do not improve on it: M2 has median −0.02 and M3 −0.14. The M3→M2 contrast — the one that isolates $(\alpha_{t-1}, x_{\min,t-1}, \dot{\alpha}_{t-1})$ and nothing else — excludes zero in 0 of 5 origins against the 3 required, with a median relative change in pinball loss at $\tau = 0.5$ of −1.5%.

Table 7: Ablation ladder over the five forecast origins 2021–2025, state level (26 units, 23–25 evaluated per origin). Cells give the median across origins of the skill score $1 - L(\text{rung})/L(\text{M0})$ against calibrated persistence, with the range over origins in brackets; the primary loss is the pinball loss at $\tau = 0.5$. The last column is the M3−M2 differential at that loss: median relative change, negative meaning that the scaling block worsens the forecast, and in parentheses the number of origins in which M3 improves on M2 by sign alone. Epidemic duration is included for completeness: it is the one target on which the lagged block is supported, and it enters the size model as a covariate. The classifier is reported separately in Table 6.

Target	M1	M2	M3	M3−M2
Size S_t	0.10 [−0.59, 0.71]	−0.02 [−0.75, 0.71]	−0.14 [−0.56, 0.71]	−1.5% (2/5)
Duration D_t	0.27 [−0.02, 0.43]	0.29 [0.06, 0.52]	0.27 [−0.06, 0.48]	−3.9% (1/5)
Peak week P_t	0.08 [−0.16, 0.26]	−0.03 [−0.69, 0.26]	0.06 [−0.55, 0.26]	+1.4% (3/5)

The block is also worst exactly where it was expected to be best. The argument for α_{t-1} is that it encodes how much of the risk sits in the tail, so any gain should concentrate at high τ; Figure 13b was drawn to test precisely that (Figure 13b). At $\tau = 0.95$ the point estimate of the M3−M2 differential is negative in *all five* origins — −24.9%, −23.7%, −0.2%, −1.6% and −18.3% of the M2 loss, median −18.3% — although with 23–25 units per origin no individual interval excludes zero. Adding $\hat{\alpha}_{t-1}$ and $\hat{x}_{\min,t-1}$ makes the upper quantile of the size forecast worse. This is what the coupling predicts: both statistics are themselves functions of the magnitude of the previous season, so at the upper tail the model is handed a second and noisier copy of S_{t-1} and weights it as if it were independent information.

Two origins dominate the size column and need separate mention. The 2024 season was the largest on record and 2025 fell back sharply; persistence extrapolated the record into 2025, and in absolute terms lost there much of what it had gained in 2024. The pinball losses for size are accordingly an order of magnitude larger in those two origins (1.5–1.7×10^5 cases) than in the other three (1.2–3.2×10^4), so a pooled mean over origins would mostly describe how each rung behaves across a regime shift. We therefore report the median across origins and the sign of the per-origin differential, not a pooled average.

That regime shift is the substantive lesson of this subsection, and it is consistent with the descriptive result rather than in tension with it. If epidemic size follows $\alpha \approx 2$, the largest season a municipality will experience is systematically outside the support of its observed history, and tree ensembles cannot extrapolate beyond their training responses. A model that already knows the previous season's size will therefore be approximately right in ordinary years, and wrong by a large factor in the exceptional one — which is exactly the pattern of the table, and of Figure 12.

3.7 Error structure and spatial output

Accuracy pooled over units is not the same as accuracy where decisions are made, so we report the error structure (Figure 14) and the map a surveillance service would actually read (Figure 15).

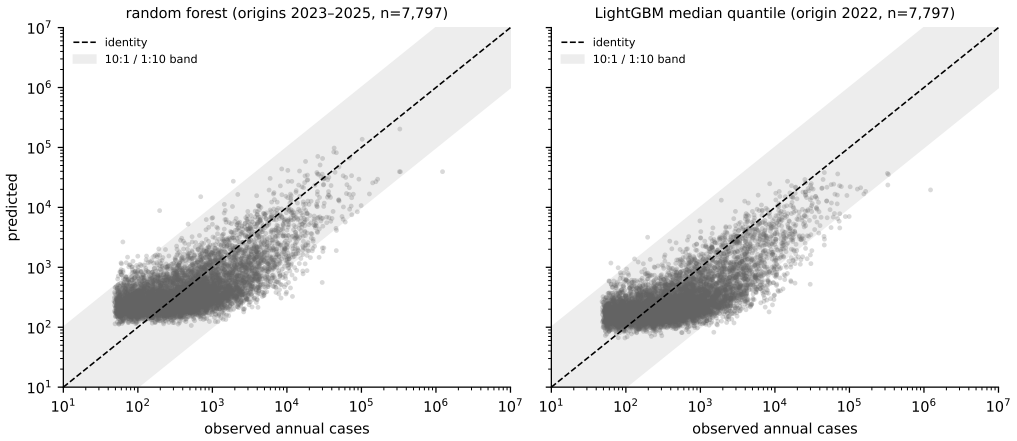


Figure 12: Out-of-sample forecasts of epidemic size. Predicted against observed annual cases on log–log axes, one point per unit and forecast origin, for the random forest (left) and the median of the gradient-boosted quantile ensemble (right); the identity line is dashed and the 10:1 and 1:10 bands are shaded. Points inside the bands are forecasts within an order of magnitude of the truth. Because $\alpha \approx 2$, a handful of points in the upper right dominate any squared-error summary, which is why the primary loss in Table 7 is the pinball loss rather than the RMSE.

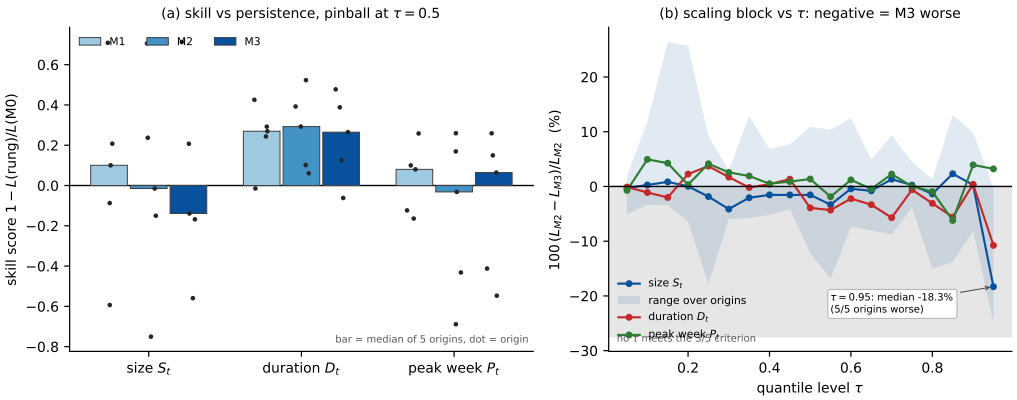


Figure 13: The ablation ladder, drawn directly from `ablation_results.csv` rather than recomputed. (a) Skill score relative to the calibrated persistence benchmark (M0) for epidemic size, duration and peak week at the primary loss (pinball, $\tau = 0.5$); bar height is the median over the five forecast origins and each dot is a single origin. The M2→M3 step isolates the contribution of the scaling descriptors. (b) The quantity the decision rule turns on, $100\,(L_{M2} - L_{M3})/L_{M2}$, against τ: a gain confined to high τ would mean the scaling information concerns extreme seasons specifically, which is what a heavy-tail description should deliver, so negative values here are the opposite of the intended effect. Lines are medians over origins and the band is the range across origins. The block degrades the extreme quantile of the size forecast, with a median loss of 18.3% at $\tau = 0.95$ and a negative differential in all five origins. Skill scores carry no interval of their own: the unit-bootstrap interval belongs to the paired differential.

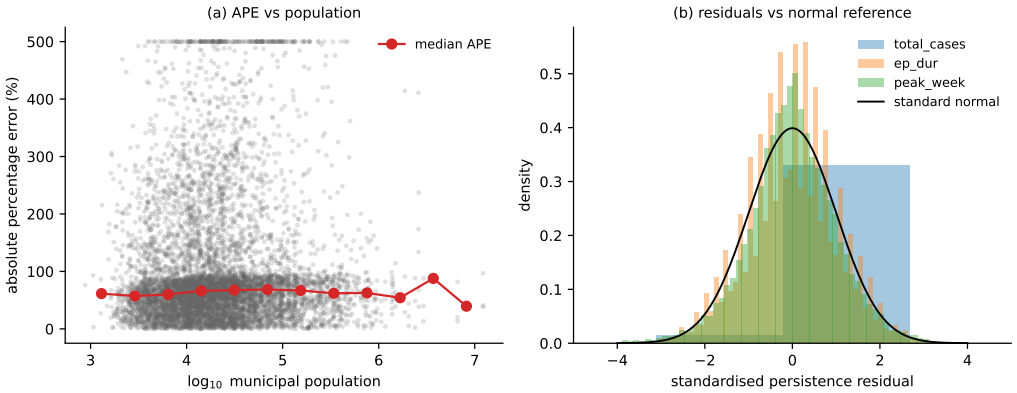


Figure 14: Forecast error against population size. Absolute percentage error of the size forecast by unit population, one point per unit and origin, with a locally weighted smooth. Errors are largest for the smallest municipalities, where case counts are low and a single imported case is a large relative change, and they narrow but do not vanish at the top of the size distribution.

4 Discussion

4.1 What the exponent is, and what it is not

The seven-level comparison settles a question that the applied literature has treated as empirical when it is largely definitional. Median exponents span 1.95 to 2.95, unanimous support for a power law at one level coexists with unanimous rejection at another, and a single change of temporal window moves support from 75% to 23% with no transmission parameter altered. The step that does not survive is the inference from an exponent to a regime: the step that treats α as a property of the pathogen or the place, when it is a property of the tuple (unit, window, variable, comparison model).

But the same comparison identifies which construction *does* carry a stable tail. Epidemic size across units within a year is power-law at every level where $x_{\min}$ lies above the bulk: $\alpha = 2.00$ in 15 of 15 national cross-sections, 96% support across 330 state-years, and a lower bound in the hundreds of cases, two to three orders of magnitude above the body of the distribution. That is not a fragile result, and it is the one worth building on. The distinction between this and the weekly-count fits that collapse to $x_{\min} \leq 5$ is not a technicality: at $x_{\min} = 2$ the fitted law describes counts of one and two cases, and at $x_{\min} = 762$ it describes the upper decade of a distribution. Two analyses reported as agreeing on the exponent can be describing entirely different objects — L1 and L3 have median exponents of 1.95 and 2.09, a difference of 0.14, while their median lower bounds differ by a factor of 170.

4.2 The tail is a risk specification, not a forecast input

The most consequential move in this paper is to stop treating $\alpha \approx 2$ as a verdict on forecastability and treat it as a probability statement. A power law assigns $\Pr(S \geq s)$ at every level; what it denies is that the *magnitude* of the largest event can be estimated from the variance. Those are different claims, and conflating them is what produced the intuition that heavy-tailed epidemics are unforecastable.

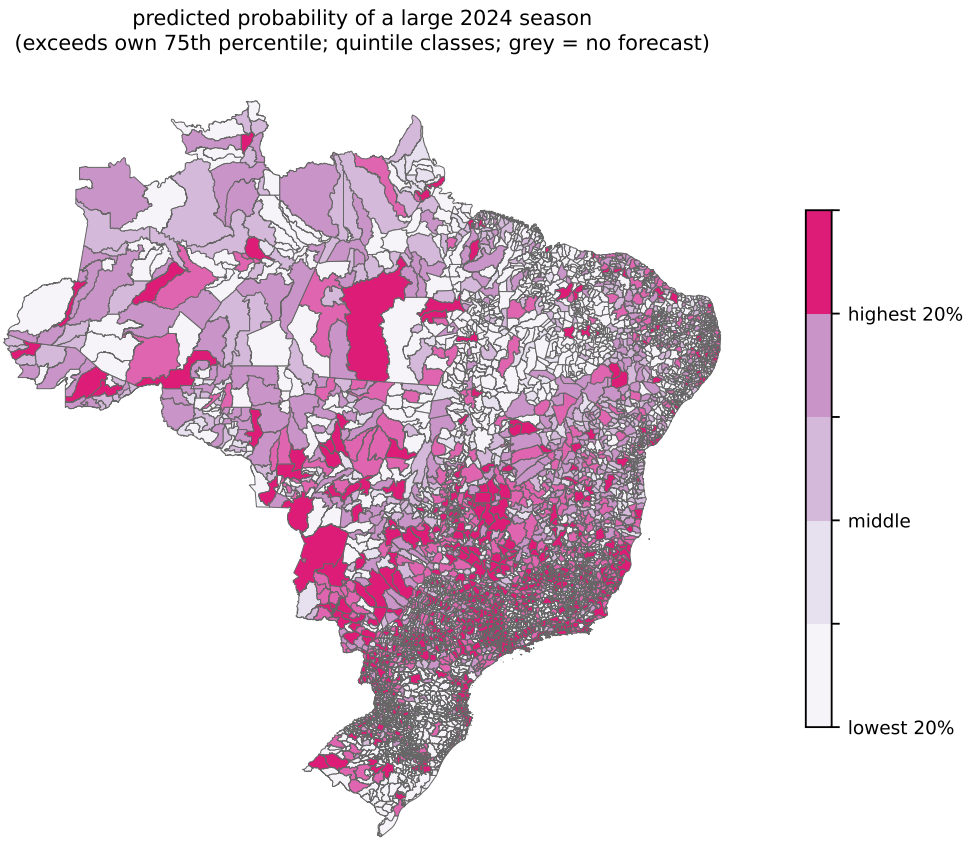


Figure 15: Out-of-sample forecasts of epidemic size by state for one forecast origin, in the form in which a surveillance service reads them. The map shows the median quantile forecast of annual cases; the legend is truncated at the 95th percentile of the plotted values so that the largest states do not flatten the colour scale for the rest.

Read as a risk specification, the fitted pair performs well and cheaply. Two parameters reproduce the empirical exceedance frequency over two decades of epidemic size (Figure 9), and they imply that between 5.4% and 16.5% of municipalities exceed ten times $x_{\min}$ in a given year against 0.3–2.7% exceeding a hundred times it. No covariate, no training period and no fitted case count is involved: the statement is available from the current cross-section, for every municipality, including those that have never experienced an extreme season. For a service deciding how many beds to hold in reserve, that is closer to the decision than a point forecast is.

The predictive evaluation confirms the division of labour. Where the question is posed as a probability — will this municipality exceed its own upper quartile? — the answer is yes, and detectably so: lagged epidemic descriptors improve the Brier score over calibrated persistence in 3 of 5 origins, by up to 0.18 (Table 6). Where the question is posed as a magnitude, the answer is no, and the failure is orderly rather than mysterious.

4.3 Why the coupling explains both results

The coupling between α and $x_{\min}$ is the mechanism behind both halves of the predictive result, and it deserves to be read as a finding rather than as a caveat. Within units the two estimates are rank-correlated at 0.65 (median over 24 states) and at 0.87 nationally, while a simulation with the true exponent held fixed at 2.0 recovers -0.20. The association is a property of the data, not of the estimator.

Descriptively, that makes the pair a compact readout of seasonal magnitude: two numbers that move with the size of the season, vary far less than the case counts they summarise (SD 0.154 against a sampling noise of 0.03), and are available for every unit. This is a real contribution to how a season can be characterised — it is a shrinkage of a two-order-of-magnitude quantity into a stable low-dimensional summary.

Predictively, the same fact is the explanation for the null. An exponent that co-varies with the scale of the season is, to that extent, a re-encoding of S_{t-1}, which the model already has. The ablation measures exactly that redundancy: the scaling block adds nothing to a rung that already knows last season's size (0 of 5 origins), and at $\tau = 0.95$ it is actively harmful (median -18.3%, negative in all five origins), because at the upper tail the model receives a second, noisier copy of S_{t-1} and weights it as independent information. The redundancy was predictable from the coupling before the ablation was run, and the ablation confirmed the prediction. That is a stronger form of evidence than a bare null: the sign and the location of the effect were specified in advance.

It also explains why the exceedance question behaves differently. Asking whether a season will be large is a question about upper-tail probability, and the coefficient on the lagged descriptors there is doing the work; adding a transformation of the same lagged quantity cannot improve it, which is why the scaling block's Brier differentials sit at ± 0.01 with no interval excluding zero. The information the exponent carries is real, and it is already in the lagged count.

4.4 Size is the hard target, and that is consistent with the description

Nothing in this paper shows that epidemic size can be forecast well. Against calibrated persistence no rung achieves skill in more than 2 of 5 origins, and two of the five origins dominate the loss because they straddle a regime shift: 2024 was the largest season on record and 2025 fell back sharply.

This is not an anomaly to be explained away; it is what $\alpha \approx 2$ implies for the estimator class we used. Tree ensembles predict within the support of their training responses, and under a heavy tail the season of greatest public-health interest is systematically outside that support. A model that already knows the previous season will be approximately right in ordinary years and wrong by a large factor in the exceptional one. The quantile models partially address this — a regression at $\tau = 0.95$ at least attempts to represent the upper region rather than interpolating within the observed range — but the limitation is structural to the method, not a matter of feature engineering.

The honest summary is therefore a division of labour rather than a verdict. The tail does not tell you how big next season will be; it tells you how likely an extreme season is, and it does so with two parameters and no training data. Those are different products for different decisions, and the second is the one a surveillance service can use before the season starts.

4.5 A reporting convention

The practical consequence of the descriptive half is a convention, and it is cheap to adopt. An exponent should travel with the lower bound that produced it, the unit of analysis, the number of observations, and the alternative model it was tested against. Of these, $x_{\min}$ is the most informative and the most frequently omitted, because it separates two analyses that look identical in a table of results. The diagnostic is easy to apply and hard to game, since $x_{\min}$ is chosen by the same goodness-of-fit criterion that selects the model and cannot be tuned to produce a desired α. We would go further: an α reported without $x_{\min}$ should be treated as an incomplete statistic, in the same way that a correlation is incomplete without its sample size.

We would add a second convention, specific to the argument this paper tests. A claim that a scaling description improves forecasts is only meaningful against a specified benchmark, on a proper scoring rule, out of sample, with the scaling block isolated from the blocks it is added to, and with every leakage channel closed. The redundancy documented here would have been invisible under a pooled R^2 against no benchmark: an earlier run of the same pipeline reported an out-of-bag R^2 of 0.35 for size, computed on the pooled cross-section on the log scale, which reflected between-unit differences that persistence reproduces trivially.

4.6 Relation to operational forecast systems

Operational arbovirus platforms — InfoDengue and Mosqlimate in Brazil, and the national systems reviewed elsewhere — already produce weekly municipal nowcasts and short-horizon forecasts (Araujo et al., 2025). Those systems answer the question this paper finds hard: what is happening now, and what will happen in the next few weeks. The scaling description answers a question they mostly do not: given this municipality's history, how often should its cross-section produce a season of a given size, expressed as a probability with two parameters. The two are complementary, and the second is cheap enough to be computed for every municipality in the country from a single cross-section.

4.7 Limitations

Several limitations bound these conclusions.

The scaling block does not improve the size forecast. We state this in the Results rather than only here because it is a measured outcome, but it is the principal negative result of

the paper and should travel with any use of it. Against a benchmark that already carries the lagged case count, the block of $(\alpha_{t-1}, x_{\min,t-1}, \dot{\alpha}_{t-1})$ fails the pre-registered test on size in 0 of 5 origins, and at the 95th percentile it degrades the loss in every origin tested, by a median of 18.3%. The mechanism is the coupling documented in Section 3.3. A rung containing the scaling descriptors *alone*, without the lagged metrics, would answer a different and interesting question — whether three power-law parameters can substitute for the full covariate stack — and is not reported here.

Epidemic duration and peak timing are not scaling objects. Both are bounded above by the length of a season, so a scale-free law cannot apply to them a priori, and the fits confirm it: the power law is rejected in 15 of 15 national duration cross-sections, with the estimated exponent piling up at 2.95 against the estimator's ceiling of 3. A near-constant fitted exponent across independent cross-sections is not evidence of universality; it is evidence that the model is being asked the wrong question. For that reason we report duration and peak timing only as covariates and as forecast targets in their own right — where, notably, duration is the one target on which the lagged block is supported (median skill 0.27, 3 of 5 origins) — and we do not build a scaling argument on them.

The case series are nowcast-adjusted. The variable analysed is the estimated case count produced by the platform's nowcasting model, not the raw notification count. This is the right series for a retrospective description, because raw notifications are incomplete in recent weeks, but it means that the upper tail inherits whatever smoothing the nowcast applies. If the nowcast shrinks extreme values, the fitted exponents are biased towards the bulk, and the size distribution reported here is somewhat less heavy than the notification distribution would be.

The state level is an arithmetic aggregate. State-level duration is the mean of municipal durations and state peak week is the median of municipal peaks. Neither is the duration or the peak of the aggregate epidemic, which is the quantity a state health secretariat actually experiences. The state-level results should be read as properties of the municipality distribution within a state, not as properties of the state epidemic.

The cross-sections are short. The national levels rest on 15 annual fits and the state levels on three to 15 years per state, so the correlations reported in Section 3.3 are estimated on small samples even where the underlying cross-sections contain thousands of municipalities. The within-state estimates are more robust than the national ones because they pool over 24 states.

The trees cannot extrapolate. This limitation cuts across both halves of the paper. Under $\alpha \approx 2$ the largest season a municipality will experience is systematically outside the support of its observed history, so the models are weakest precisely where the heavy tail places the risk. No amount of feature engineering removes that; it is a property of the estimator class.

The classification threshold is a choice. Exceeding the 75th percentile of a unit's own history defines a large season; at the 90th percentile the positive class becomes small enough that the balanced accuracy of any classifier falls and the comparison between rungs loses power. The threshold is therefore a design parameter, fixed before the runs and reported with them.

The exponent cannot exceed three. The fitting routine bounds α above by 3, so a tail steeper than that is reported as 3 rather than as its unconstrained maximum. For the size results this is immaterial, since no national size fit exceeds 2.27; for durations it is not, and 11 of the 330 state-year fits sit at the bound. This does not weaken the conclusion that durations are not power-law distributed, but the numerical value of those particular exponents should not be read as an estimate.

Predictors are ecological and coarse. No serotype composition, no seroprevalence and no vector index enters the model, although all three are plausible determinants of season size. To the extent that these are stable within a unit, they are absorbed by unit-level effects; to the extent that they vary by year, they are unmodelled heterogeneity that the scaling block may be inadvertently proxying. This is a live hypothesis for the null reported here, and it is the one we would test first.

4.8 Implications for surveillance practice

Three practices follow, none of them expensive.

First, report the tuple. An exponent accompanied by its unit, its $x_{\min}$, its sample size and its comparison model can be assessed by a reader; an exponent alone cannot, and the seven levels here show that the same exponent can mean opposite things.

Second, use the exceedance curve. Two parameters fitted to the current cross-section give the probability that a municipality will exceed any level, and the national fits reproduce the observed frequency of large seasons over two decades. This is a cheap input to planning that requires no model training and no history beyond the present year.

Third, do not translate a heavy tail into a claim of unpredictability without specifying the loss. A distribution without a finite variance constrains squared-error point forecasts and leaves probabilistic exceedance statements intact — and the exceedance statement is usually the one a health service needs.

5 Conclusion

The distribution of dengue epidemic size in Brazil is heavy-tailed with an exponent close to two, and this is one of the few robust facts in the applied literature on epidemic scaling: 15 of 15 national cross-sections, 96% of 330 state-years, and a lower bound two to three orders of magnitude above the bulk. Which of these statements a study reports, and whether it reports a power law or an exponential at all, is determined by the construction it imposes on the data; that heterogeneity is methodological before it is biological.

The consequence for forecasting is not the one usually drawn. Because $\alpha \approx 2$ fixes the entire exceedance curve, the fitted pair is a two-parameter risk specification, and it is well calibrated against the empirical frequency of large seasons over two decades of epidemic size. Posed as a probability — will this municipality exceed its own upper quartile? — the problem is tractable: lagged epidemic descriptors improve the Brier score over calibrated persistence in 3 of 5 forecast origins. Posed as a magnitude, it is not: no rung achieves skill for size, and the scaling block specifically adds nothing (0 of 5) while degrading the upper quantile by a median 18%.

That last result is not an accident of the data. The exponent co-varies with the lower bound and hence with the scale of the season, at 0.87 nationally and 0.65 within states, in a way that simulation does not reproduce. The pair is therefore a genuine low-variance summary of how large a season was, and for that same reason a redundant input to any model that already knows how large the previous one was. The redundancy was predicted from the coupling before the ablation was run, and the ablation confirmed it.

What generalises beyond this dataset is the shape of the question. A heavy tail does not make epidemics unpredictable; it bounds how well the magnitude of the largest event can be estimated from second moments, and it leaves the probability of exceeding any level fully specified. The exponent is worth reporting. It is worth reporting as a risk curve, with its lower bound, and not as a predictor.

A Verification of reported quantities

Every quantity in Table 4 is obtained from `powerlaw_results.db` by the queries in `Manuscript/verification/`. The comparison columns are computed as

`R>0` ⇒ PL preferred; `R>0 AND p<0.05` ⇒ PL supported; `R<0 AND p<0.05` ⇒ exponential supported.

Note that `year = 0` in the `episcanner` tables is a pooled all-years fit, not a calendar year, and is excluded from all year counts; the national levels therefore have 15 annual fits, not 16.

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